RESEARCH POSTER / MULTI-SCALE NETWORK SECURITY

IDS-MTRAN

A Transformer-based intrusion detector that examines network traffic at three different zoom levels.

99%+
reported accuracy
on all three datasets

58.61
traffic samples/sec
average test speed

Xi, Wang & Wang | Scientific Reports 2024, 14:23239 | DOI: 10.1038/s41598-024-74214-w

What exactly is this work about?

IDS-MTran is a smart network guard. Instead of examining traffic in only one way, it studies the same data at three zoom levels: fine details, middle-sized patterns and broad trends. Transformers study each view, and the model combines them before deciding whether the traffic is normal or an attack.

HIDDEN ATTACK SIGNAL
small clue in busy traffic

The key idea: different attacks leave clues at different levels, so one view may not be enough.

→

THREE-SCALE VIEW
detail + pattern + trend

01 WHAT THE SYSTEM RECEIVES

NETWORK TRAFFIC

Connection and packet-related features

The records describe services, protocols, flags, timing, source, destination and other traffic properties.

DATASET 1

NSL-KDD

148,517 processed samples across normal traffic and four attack groups.

DATASET 2

CIC-DDoS 2019

A very large, highly unbalanced collection focused on DDoS attacks.

DATASET 3

UNSW-NB15

2,540,044 realistic records covering modern traffic and attack types.

Two tasks are evaluated. Binary classification asks normal or malicious. Multi-class classification also identifies the particular attack type.

02 HOW THE SYSTEM WORKS

1 Prepare the traffic

Text values become numbers, extreme values are handled, and every feature is put on a similar scale.

2 Create three views

Different filters produce fine, medium and broad views of the same traffic record.

3 Reduce noise and make patches

PwP highlights useful information, lowers noise and divides each view into manageable pieces.

4 Study each view

A Transformer examines relationships between distant parts of the traffic data.

5 Share information

CFE lets the three branches exchange details so each view becomes richer.

6 Make the final decision

The combined information is used to predict normal traffic or a specific attack type.

1 Traffic → 2 3 views → 3 Patches → 4 Attention → 5 Decision

06 WHY THREE SCALES HELP

Fine view — Finds short, local changes such as sudden request bursts or packet-size variation.

Broad view — Recognizes longer trends and the overall behaviour of a traffic connection.

Middle view — Connects local clues with broader patterns so neither perspective is lost.

How the model handles rare attacks

1. Real traffic datasets contain many common records and relatively few rare attacks.
2. Normal training can focus too much on easy, common examples.
3. Focal Loss gives extra attention to difficult and less-common examples during learning.

03 WHAT THE SYSTEM GIVES

FIRST DECISION

Normal or Malicious

A quick binary result for each traffic record.

DETAILED DECISION

Specific attack type

For example: DoS, Probe, U2R, R2L or a DDoS category.

QUALITY MEASURES

Accuracy, precision, recall and F1

The paper reports both overall and class-by-class performance.

OPERATIONAL USE

Alert for an administrator

The predicted threat can be logged and sent for investigation.

Practical purpose. Detect both obvious and subtle attack patterns in busy network traffic, including difficult minority attack categories.

04 RESULTS REPORTED IN THE PAPER

The tasks differ by dataset. Values below are author-reported test results.

DATASET / TASK	ACCURACY	PRECISION	RECALL	F1
NSL-KDD / binary	**99.25%**	**99.07%**	**99.02%**	**99.05%**
NSL-KDD / 5 classes	**99.16%**	**99.01%**	**99.17%**	**99.09%**
CIC-DDoS / multi-class	**99.07%**	**99.81%**	**99.42%**	**99.61%**
UNSW-NB15 / multi-class	**99.70%**	**98.50%**	**99.80%**	**99.10%**

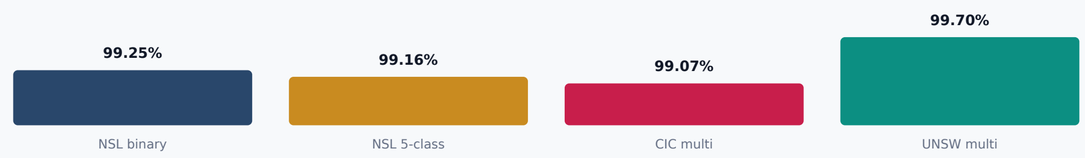
58.61
traffic samples per second
average on an RTX 4090

3 scales
fine, middle and broad views
are combined

100
maximum training epochs
with early stopping

1.24-2.76
accuracy points gained by
cross-feature enrichment

Accuracy across the reported tasks



Read the scores carefully

These are results on three prepared benchmark datasets, not a live network. The paper does not clearly describe the train/test split procedure, and the speed was measured on a powerful RTX 4090 workstation. Independent reproduction is still needed.

05 WHERE IT CAN BE USED

- Security operations centres. Rank and classify suspicious traffic for analyst investigation.
- DDoS monitoring. Recognize multiple denial-of-service attack patterns in large traffic streams.
- Multi-attack classification. Identify not only that an attack occurred, but also its likely category.

What the results actually show. Looking at traffic at several scales improves attack-type recognition on the selected benchmarks. The model is more accurate than the compared neural baselines, but it is slower than simpler CNN, RNN and LSTM models.

07 CHECKS NEEDED BEFORE REAL USE

Future traffic

Train on earlier traffic and test on later periods to detect dataset drift.

Unseen attacks

Hold out complete attack families and check whether the model can find them.

Smaller hardware

Measure speed and memory on ordinary servers, edge devices and campus networks.

Live false alarms

Test real traffic and record how many normal connections are incorrectly flagged.

Xi, Wang and Wang, 'A novel multi-scale network intrusion detection model with transformer,' Scientific Reports 14 (2024), 23239. DOI: 10.1038/s41598-024-74214-w. Values follow the paper and are author-reported, not independently reproduced.